

Zero Jordan product determination for discrete and SIN group algebras

Candidate substantial partial solution for arXiv:1902.04846

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Result and relation to the source question

Alaminos, Brešar, Extremera, and Villena prove that $L^1(G)$ is zero Jordan product determined when G is amenable, and ask whether this holds for every locally compact group [1, Question 3.7]:

Theorem 3.6. *The following Banach algebras are zero Jordan product determined:*

- (1) *the group algebra $L^1(G)$ of an amenable locally compact group;*
- (2) *every C^* -algebra.*

Question 3.7. *Is the group algebra $L^1(G)$ a zero Jordan product determined Banach algebra for each locally compact group G ?*

We will now provide an example of a noncommutative Banach algebra

Recall that a locally compact group is a *SIN group* if the identity has a neighborhood basis invariant under conjugation.

Theorem 1. *If G is a locally compact SIN group, then $L^1(G)$ is zero Jordan product determined. In particular, $\ell^1(G)$ is zero Jordan product determined for every discrete group G .*

Thus there is no amenability, torsion, or finite-generation restriction in the discrete case. The result includes the torsion-free nonamenable free groups. It also includes nondiscrete nonamenable groups, for example $F_2 \times \mathbb{R}$.

The commutator criterion

For a Banach algebra A , put $a \circ b = ab + ba$ and

$$\mathcal{J}(A) = \overline{\text{span}}\{ab : a, b \in A, a \circ b = 0\}.$$

The source proves the following criterion: if A is weakly amenable, has property $\mathbb{B}$, and has a bounded approximate identity, then A is zero Jordan product determined if and only if

$$[A, A] \subseteq \mathcal{J}(A). \tag{1}$$

It also records that every group algebra $L^1(G)$ has all three hypotheses [1]. We will therefore prove (1).

The engine is an elementary synthesis statement for the Wiener algebra

$$\mathcal{W} = A(\mathbb{T}) = \left\{ f(z) = \sum_{n \in \mathbb{Z}} \widehat{f}(n) z^n : \|f\|_{\mathcal{W}} = \sum_n |\widehat{f}(n)| < \infty \right\}.$$

Via Fourier coefficients, $A(\mathbb{T}^2) = \mathcal{W} \widehat{\otimes}_{\pi} \mathcal{W}$ isometrically.

A separated-support Wiener lemma

Lemma 2. *In $A(\mathbb{T}^2)$ the function*

$$D(z, w) = z - w$$

belongs to the closed linear span of the tensors $g \otimes h$ with $g, h \in \mathcal{W}$ and $\text{supp } g \cap \text{supp } h = \emptyset$.

Proof. We give the cutoff proof in full. For small $\varepsilon > 0$, choose a smooth 2π -periodic function χ_ε which is 1 for $|t| \leq \varepsilon$, is 0 for $|t| \geq 2\varepsilon$, and satisfies $|\chi'_\varepsilon| \leq C/\varepsilon$. Define

$$d_\varepsilon(e^{it}) = (e^{it} - 1)\chi_\varepsilon(t), \quad q_\varepsilon(z) = (z - 1) - d_\varepsilon(z).$$

For every smooth d on $\mathbb{T}$, Cauchy–Schwarz on its Fourier coefficients gives

$$\|d\|_{\mathcal{W}} \leq \left(\sum_{n \in \mathbb{Z}} (1 + n^2)^{-1} \right)^{1/2} \|d\|_{H^1(\mathbb{T})}. \quad (2)$$

On an arc of length $O(\varepsilon)$ we have $|e^{it} - 1| = O(\varepsilon)$. Hence

$$\|d_\varepsilon\|_2 = O(\varepsilon^{3/2}), \quad \|d'_\varepsilon\|_2 = O(\varepsilon^{1/2}),$$

and (2) implies $\|d_\varepsilon\|_{\mathcal{W}} \rightarrow 0$.

Write $r = z\bar{w}$. Since $D(z, w) = w(r - 1)$, the smooth function

$$Q_\varepsilon(z, w) = wq_\varepsilon(z\bar{w})$$

vanishes on a neighborhood of the diagonal. Moreover the change of Fourier indices ($n \mapsto (n, 1 - n)$) gives

$$\|D - Q_\varepsilon\|_{A(\mathbb{T}^2)} = \|d_\varepsilon\|_{\mathcal{W}} \rightarrow 0. \quad (3)$$

It remains to decompose Q_ε . Its compact support is covered by finitely many product arcs $U_j \times V_j$ with $\bar{U}_j \cap \bar{V}_j = \emptyset$. Take a smooth partition of unity on the support and write $Q_\varepsilon = \sum_j Q_j$, where $\text{supp } Q_j \subseteq U_j \times V_j$. Choose smooth α_j, β_j supported in U_j, V_j , respectively, and equal to 1 on the corresponding coordinate projections of $\text{supp } Q_j$. The double Fourier series

$$Q_j(z, w) = \sum_{m, n \in \mathbb{Z}} c_{mn}^{(j)} z^m w^n$$

has absolutely summable coefficients. Therefore, in projective norm,

$$Q_j = \alpha_j(z)\beta_j(w)Q_j(z, w) = \sum_{m, n} c_{mn}^{(j)} (\alpha_j(z)z^m) \otimes (\beta_j(w)w^n). \quad (4)$$

The two factors in every tensor in (4) have disjoint supports. Equations (3)–(4) prove the lemma. $\square$

All discrete groups

Let G be any discrete group and $s \in G$. There is a contractive unital homomorphism

$$\pi_s : \mathcal{W} \longrightarrow \ell^1(G), \quad \pi_s(f) = \sum_{n \in \mathbb{Z}} \hat{f}(n) \delta_{s^n}. \quad (5)$$

This remains valid when s has finite order: repeated point masses simply combine, and absolute convergence proves both contractivity and multiplicativity.

Fix $s, t \in G$ and define a bounded map

$$\Theta_{s,t} : A(\mathbb{T}^2) \longrightarrow \ell^1(G), \quad \Theta_{s,t}(g \otimes h) = \pi_s(g) \delta_t \pi_s(h). \quad (6)$$

If $\text{supp } g \cap \text{supp } h = \emptyset$, choose $u \in C^\infty(\mathbb{T})$ such that $u = 1$ on $\text{supp } g$ and $u = -1$ on $\text{supp } h$. Set

$$a = \pi_s(u), \quad b = \pi_s(g) \delta_t \pi_s(h).$$

Since $ug = g$ and $hu = -h$, we have

$$ab = b, \quad ba = -b.$$

Thus $a \circ b = 0$ and $b = ab \in \mathcal{J}(\ell^1(G))$. Lemma 2, the continuity of (6), and

$$\Theta_{s,t}(D) = \delta_s \delta_t - \delta_t \delta_s$$

show that every point-mass commutator belongs to $\mathcal{J}(\ell^1(G))$.

For arbitrary $p, q \in \ell^1(G)$,

$$[p, q] = \sum_{s, t \in G} p(s)q(t) [\delta_s, \delta_t] \quad (7)$$

converges absolutely in $\ell^1(G)$, since the sum of the norms is at most $2\|p\|_1\|q\|_1$. The space $\mathcal{J}(\ell^1(G))$ is closed and linear, so (7) proves (1). The source criterion completes the discrete case.

Transfer to locally compact SIN groups

Let now G be locally compact and SIN. Shrinking relatively compact conjugation-invariant neighborhoods U of the identity give

$$e_U = m(U)^{-1} \mathbf{1}_U \in L^1(G). \quad (8)$$

These form a contractive approximate identity. They are central even in the measure algebra $M(G)$. Indeed, invariance of one such U of finite positive measure shows that inner automorphisms preserve Haar measure, so G is unimodular; conjugation invariance then gives $\delta_s * e_U = e_U * \delta_s$ for every s , and integration gives commutation with every measure.

Formula (5) now defines a contractive homomorphism $\pi_s : \mathcal{W} \rightarrow M(G)$. Fix $s \in G$ and $k \in L^1(G)$, and put

$$\Theta_{s,k}(g \otimes h) = \pi_s(g) * k * \pi_s(h). \quad (9)$$

For a separated-support tensor, choose u as above and set

$$a = \pi_s(u) \in M(G), \quad b = \pi_s(g) * k * \pi_s(h) \in L^1(G).$$

Again $a * b = b$ and $b * a = -b$. Although a need not lie in $L^1(G)$, define $a_U = a * e_U \in L^1(G)$. Centrality of e_U gives the *exact* identities

$$a_U * b = b * e_U, \quad b * a_U = -b * e_U,$$

so $a_U \circ b = 0$ and $b * e_U = a_U * b \in \mathcal{J}(L^1(G))$. Letting U shrink yields $b * e_U \rightarrow b$, hence $b \in \mathcal{J}(L^1(G))$. Applying Lemma 2 to (9) now gives

$$[\delta_s, k] = \delta_s * k - k * \delta_s \in \mathcal{J}(L^1(G)) \quad (10)$$

for all $s \in G$ and $k \in L^1(G)$.

If $f \in C_c(G)$, norm continuity of left and right translation makes the following a Bochner integral in $L^1(G)$:

$$[f, k] = \int_G f(s) [\delta_s, k] ds. \quad (11)$$

The integrand in (11) takes values in the closed linear space $\mathcal{J}(L^1(G))$ by (10), so its Bochner integral does too. Density of $C_c(G)$ in $L^1(G)$ extends this to every $f \in L^1(G)$. Thus (1) holds, and the source criterion proves Theorem 1.

Boundary of the method and novelty status

For a general locally compact group the cyclic calculus (5) still lives in $M(G)$, but a bounded approximate identity need not be central. Replacing a by $a * e_i$ then destroys the exact relation $(a * e_i) \circ b = 0$. Approximate anticommutation is insufficient for the definition of $\mathcal{J}(A)$. We therefore do not claim a solution for arbitrary locally compact groups.

A bounded novelty search on 27 August 2026 checked the run indexes and close web/arXiv combinations of the source id with “all discrete groups”, “SIN group”, “central approximate identity”, “Wiener algebra”, and “diagonal synthesis”. No explicit statement of either theorem above was found. This is provisional: an expert citation search is still required.

Human review should focus on projective-norm convergence in (4), the central approximate identity in (8), and the multiplier-to- L^1 transfer. These steps are included explicitly to make the candidate independently auditable.

References

- [1] J. Alaminos, M. Brešar, J. Extremera, and A. R. Villena, *Zero Jordan product determined Banach algebras*, J. Aust. Math. Soc. 111 (2021), 145–158; arXiv:1902.04846.